

# Raphaël Tinarrage

## Curriculum vitae

Institute of Science and Technology Austria  
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🌐 [raphaeltinarrage.github.io](https://github.com/raphaeltinarrage)



Links: [ORCID](#) — [ResearchGate](#) — [Google Scholar](#) — [arXiv](#) — [HAL](#) — [theses.fr](#) — [Lattes](#) — [GitHub](#) — [YouTube](#)

## Academic positions

- 2024–present **Postdoc**, Institute of Science and Technology Austria (ISTA), Klosterneuburg  
Classifying spaces in Topological Data Analysis, in Uli Wagner's group
- 2021–2024 **Postdoc**, Fundação Getulio Vargas – Escola de Matemática Aplicada (FGV EMAp), Rio de Janeiro  
Theory and applications of Topological Data Analysis, supervised by César Camacho

## Education

- 2017–2020 **PhD**, Inria Saclay and Laboratoire de Mathématiques d'Orsay  
Topological inference from measures and vector bundles — [HAL:tel-02970491](#)  
Supervised by Frédéric Chazal and Marc Glisse  
[Manuscript](#) — [slides](#) — [reports](#)
- 2014–2017 **Master's degree**, Université Paris-Saclay  
M2 research – Mathematics for life sciences  
M1 – Fundamental and applied mathematics & Magistère de mathématiques 2<sup>nd</sup> year
- 2015–2016 **Master's degree**, École Normale Supérieure Paris-Saclay  
M2 FESUP – Preparation for the *agrégation* examination (teaching diploma)
- 2013–2014 **Bachelor's degree**, Université Paris-Saclay  
L3 – Fundamental and applied mathematics & Magistère de mathématiques 1<sup>st</sup> year
- 2011–2013 **Classes préparatoires**, Lycée Camille Pissarro, Pontoise  
MPSI & MP

## Examinations, competitions, fellowships

- 09/2024 **ISTA-Fellow: Postdoctoral Program**, Institute of Science and Technology Austria (ISTA)  
Two-year fellowship
- 11/2023 **Competition for Professor Adjunto**, Universidade do Estado do Rio de Janeiro (UERJ)  
[Rank: 1<sup>st</sup>](#) (out of [10 participants](#))
- 07/2016 **Agrégation externe de mathématiques**, French teaching diploma  
[Rank: 68<sup>th</sup>](#) (out of 1896 participants)

## Teaching

- 2024 **Vector Calculus**, FGV EMAp, Rio de Janeiro  
2<sup>nd</sup> year undergraduate course (30 hours)  
[Course webpage](#) — [notes](#) (original document, 180 pages, in Portuguese)
- 2023 **General and Combinatorial Topology**, FGV EMAp, Rio de Janeiro  
Summer course for undergraduate and master's students (26 hours)  
[Course webpage](#) — [notes](#) (original document, 95 pages, in English)
- 2021 **Topological Data Analysis with Persistent Homology**, FGV EMAp, Rio de Janeiro  
Summer course for undergraduate and master's students (22 hours)  
[Course webpage](#) — [notes](#) (original document, 97 pages, in English) — [videos](#)
- 2017–2020 **Statistical Interpretation of Data**, UE M331, L3 MINT, Université Paris-Saclay, Orsay  
Teaching fellow for undergraduate students
- 2017–2020 **Modeling Project**, UE M326, L3 MINT, Université Paris-Saclay, Orsay  
Teaching fellow for undergraduate students
- 2017–2019 **Ordinary Differential Equations**, UE M257, L2 BC, Université Paris-Saclay, Orsay  
Teaching fellow for undergraduate students  
[Course webpage](#)

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## Mentoring

- 2022–2023 **Fine-tuning legal language models via annotations**, FGV EMap, Rio de Janeiro  
BSc students: Livia Cales, Victoria Cury, Samanta Duarte, Clara Lopes, Eduardo Portol, João Meirelles, Ana Rosenburg, and Helena Torres
- 2021–2023 **Data analysis of symmetries with Lie groups**, FGV EMap, Rio de Janeiro  
MSc student: Henrique Ennes
- 2021–2023 **Machine learning for Súmulas Vinculantes**, FGV EMap, Rio de Janeiro  
BSc students: Beatriz Sabdin Chagas, Carla Marcondes Damian, Ana Clara Macedo Jaccoud, and Pedro Burlini de Oliveira

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## Academic visits

- 11/2025 **Anthea Monod Group**, Imperial College, London  
Duration: 3 days.
- 10/2022 **Grupo de Bases de Dados e de Imagens (GBdl)**, Instituto de Ciências Matemáticas e de Computação (ICMC), Universidade de São Paulo (USP), São Carlos  
Visualization of temporal graphs with zigzag persistent homology  
Host: Agma Juci Machado Traina. Duration: 1 week.

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## Academic service

- 2026–present **Organizer** of the ABCDEFG seminar, ISTA
- 2025–present **Organizer** of the Combinatorics, geometry and topology seminar (CGT), ISTA — [webpage](#)
- 2018–2025 **Reviewer** for Symposium on Computational Geometry (SoCG)
- 2025 **Reviewer** for Journal of Mathematical Imaging and Vision
- 2023 **Discussant** at the workshop Transforming the Role of International Courts and Tribunals in a New Era of Adjudication, *FGV Jean Monnet Centre of Excellence*, FGV, Rio de Janeiro — [webpage](#)
- 2023 **Reviewer** for Foundations for Undergraduate Research in Mathematics (FURM)
- 2023 **Reviewer** for SIAM Journal on Applied Algebra and Geometry (SIAGA)
- 2022 **Reviewer** for MathSciNet (American Mathematical Society)
- 2017–2020 **Organizer** of the Workshop MATH.en.JEANS, Collège Alain Fournier, Orsay  
Mathematics outreach in college — [webpage](#)

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## Journal articles

- 05/2026 **Train-Free Segmentation in MRI with Cubical Persistent Homology**  
Anton François, **Raphaël Tinarrage**  
Journal of Mathematical Imaging and Vision **68**, 20 (2026)  
[DOI:10.1007/s10851-026-01300-1](#) — [arXiv:2401.01160](#)  
22 double column pages, in English
- 09/2025 **LieDetect: Detection of representation orbits of compact Lie groups from point clouds**  
Henrique Ennes, **Raphaël Tinarrage**  
Foundations of Computational Mathematics (2025)  
[DOI:10.1007/s10208-025-09728-4](#) — [arXiv:2309.03086](#)  
136 pages, in English
- 05/2025 **Empirical analysis of binding precedent efficiency in Brazilian Supreme Court via case classification**  
**Raphaël Tinarrage**, Henrique Ennes, Lucas Resck, Lucas T. Gomes, Jean R. Ponciano, Jorge Poco  
Artificial Intelligence and Law (2025)  
[DOI:10.1007/s10506-025-09458-6](#) — [arXiv:2407.07004](#)  
67 pages, in English
- 01/2025 **ZigzagNetVis: Suggesting temporal resolutions for graph visualization using zigzag persistence**  
**Raphaël Tinarrage**, Jean Ponciano, Cláudio Linhares, Agma Traina, Jorge Poco  
IEEE Transactions on Visualization and Computer Graphics **31**, 1–18 (2025)  
[DOI:10.1109/TVCG.2025.3528197](#) — [arXiv:2304.03828](#)  
18+7 (supplementary) double column pages, in English
- 02/2023 **Recovering the homology of immersed manifolds**  
**Raphaël Tinarrage**  
Discrete & Computational Geometry **69**, 659–744 (2023)  
[DOI:10.1007/s00454-022-00409-5](#) — [arXiv:1912.03033](#)  
86 pages, in English

11/2021 **Computing persistent Stiefel-Whitney classes of line bundles**

**Raphaël Tinarrage**

Journal of Applied and Computational Topology **6**, 65–125 (2022)

[DOI:10.1007/s41468-021-00080-4](https://doi.org/10.1007/s41468-021-00080-4) — [arXiv:2005.12543](https://arxiv.org/abs/2005.12543)

61 pages, in English

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## Conference articles

06/2026 **Simplicial approximation to CW complexes with spherical Delaunay triangulations**

**Raphaël Tinarrage**

To appear in the proceedings of the [Symposium on Computational Geometry \(2026\)](#) — [arXiv:2112.07573](https://arxiv.org/abs/2112.07573)

53 pages, in English

09/2022 **O impacto da Súmula Vinculante 26 na diminuição de demanda similar no STF: uma análise quantitativa por modelos de ML**

Beatriz Sabdin Chagas, Carla Marcondes Damian, **Raphaël Tinarrage**

XI Encontro Internacional do CONPEDI Chile, 82–103 (2022)

[ISBN:978-65-5648-559-1](#)

22 pages, in Portuguese

09/2022 **Progressão de regime em crimes hediondos no Supremo Tribunal Federal: uma análise empírica pela Súmula Vinculante 26**

Ana Clara Macedo Jaccoud, Pedro Burlini de Oliveira, **Raphaël Tinarrage**

XI Encontro Internacional do CONPEDI Chile, 399–418 (2022)

[ISBN:978-65-5648-569-0](#)

29 pages, in Portuguese

06/2020 **DTM-based filtrations**

Hirokazu Anai, Frédéric Chazal, Marc Glisse, Yuichi Ike, Hiroya Inakoshi, **Raphaël Tinarrage**, Yuhei Umeda

Symposium on Computational Geometry (2019) — [DOI:LIPICs.SoCG.2019.58](https://doi.org/10.1007/978-3-030-43408-3_2)

Abel Symposia 15, Springer, Cham. (2020) — [DOI:10.1007/978-3-030-43408-3\\_2](https://doi.org/10.1007/978-3-030-43408-3_2)

[arXiv:1811.04757](https://arxiv.org/abs/1811.04757)

33 pages, in English

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## Posters

06/2022 **Simplicial approximation to CW-complexes in practice**, ATMCS, University of Oxford

[Poster](#)

06/2018 **DTM-filtrations**, Algebraic Topology: Methods, Computation and Science, IST Austria

[Poster](#)

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## Selected talks

11/2025 **London-Oxford-Paris TDA Seminar**, City St George's

Simplicial approximation, Delaunay triangulations, and the list homomorphism problem

[Slides](#) — [program](#)

06/2025 **Applied Algebraic Topology Research Network**, Online

Detection of representation orbits of compact Lie groups from point clouds

[Slides](#) — [video](#)

02/2025 **Infinite-dimensional Geometry Conference**, Erwin Schrödinger Institute (ESI)

Train-Free Segmentation in MRI with Cubical PH

[Slides](#)

07/2024 **XXIII Encontro Brasileiro De Topologia**, Universidade Federal da Bahia (UFBA)

Classifying spaces in TDA

[Slides](#) — [program](#)

01/2023 **Workshop On Legal Digital Intelligence**, FGV EMAp

TDA and Súmulas Vinculantes

[Slides](#)

11/2022 **Seminário ICMC**, Universidade de São Paulo (USP), São Carlos

TDA para escolha de resolução temporal na visualização de grafos

[Slides](#)

01/2023 **Summer School on Data Science**, FGV EMAp

I. From Topology to Data Analysis: [Slides](#) — [video](#)

II. Homological inference: [Slides](#) — [video](#)

III. Persistent Homology: [Slides](#) — [video](#)

12/2020 **Applied Algebraic Topology Research Network**, Online

Persistent Stiefel-Whitney classes

[Slides](#) — [video](#)